

Bluestock Mutual Fund Analytics Capstone Project

MUTUAL FUND ANALYTICS PLATFORM

Submitted By

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Institution

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Technologies Used

Python, Pandas, NumPy, SQLite, SQLAlchemy, SciPy, Jupyter Notebook, Power BI, GitHub

Executive Summary

The Mutual Fund Analytics Platform is an end-to-end financial analytics solution developed to analyze mutual fund performance, investor behavior, portfolio risk, and market trends. The project combines data engineering, database management, advanced analytics, and business intelligence visualization into a unified platform.

The system processes more than 46,000 NAV records and 32,000 investor transactions. Multiple financial performance metrics including CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, VaR, CVaR, and Maximum Drawdown were calculated to evaluate mutual fund performance.

A Power BI dashboard was developed to provide interactive insights into industry trends, fund performance, investor behavior, and SIP growth patterns. The project demonstrates the complete lifecycle of a financial analytics application from raw data ingestion to business intelligence reporting.

PROBLEM STATEMENT & OBJECTIVES

Problem Statement

Investors often face challenges in selecting suitable mutual funds due to the large number of available schemes, varying risk profiles, and inconsistent performance measures.

Traditional investment analysis often focuses only on returns while ignoring risk-adjusted performance, investor behavior, and portfolio diversification.

This project aims to provide a comprehensive mutual fund analytics platform that enables data-driven investment decision making.

Project Objectives

1. Collect and process mutual fund datasets.
2. Build an automated ETL pipeline.
3. Design a relational database for analytics.
4. Analyze fund performance using financial metrics.
5. Evaluate portfolio risk using quantitative techniques.
6. Study investor transaction behavior.
7. Create an interactive business intelligence dashboard.
8. Develop a fund recommendation system.

DATASET DESCRIPTION

Datasets Used

The project utilizes ten datasets.

1. Fund Master

Contains scheme details including AMFI code, fund house, category, risk grade, and plan information.

2. NAV History

Daily Net Asset Value history for all schemes.

3. AUM by Fund House

Assets Under Management statistics across mutual fund companies.

4. Monthly SIP Inflows

Monthly SIP contribution data.

5. Category Inflows

Category-wise net inflow and outflow information.

6. Industry Folio Count

Investor account statistics across the industry.

7. Scheme Performance

Fund returns, expense ratios, and risk measures.

8. Investor Transactions

Transaction-level SIP, lump sum, and redemption records.

9. Portfolio Holdings

Fund portfolio allocations and sector weights.

10. Benchmark Indices

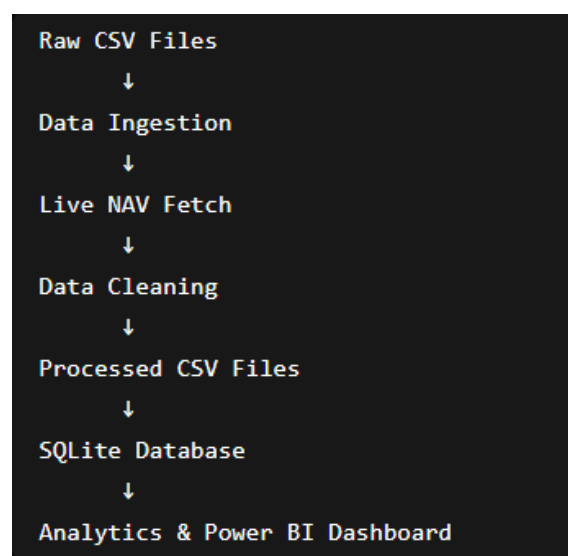
Historical benchmark index performance data.

Total Records Processed: More than 85,000 records.

DATA ENGINEERING & ETL PIPELINE

ETL Pipeline

An automated ETL pipeline was created to process mutual fund datasets.



Extraction

- Raw CSV datasets imported.
- Live NAV data fetched using MFAPI.
- Data validation performed.

Transformation

- Missing values handled.
- Duplicate records removed.
- Date formats standardized.
- Transaction categories normalized.
- Risk metrics validated.

Loading

- Cleaned datasets loaded into SQLite database.
- Star schema implemented for analytical processing.

Technologies Used

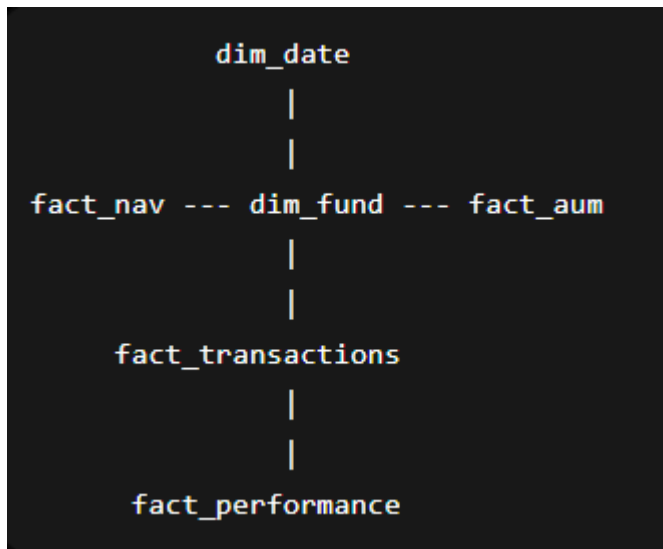
- Python
- Pandas
- Requests
- SQLAlchemy
- SQLite

The ETL pipeline executes automatically and processes all datasets without manual intervention.

DATABASE DESIGN

Database Architecture

A star schema was designed to support analytical workloads.



Dimension Tables

`dim_fund`

Stores scheme metadata.

`dim_date`

Stores calendar attributes.

Fact Tables

`fact_nav`

Daily NAV values.

`fact_transactions`

Investor transaction records.

`fact_performance`

Performance metrics.

`fact_aum`

Assets Under Management data.

Benefits

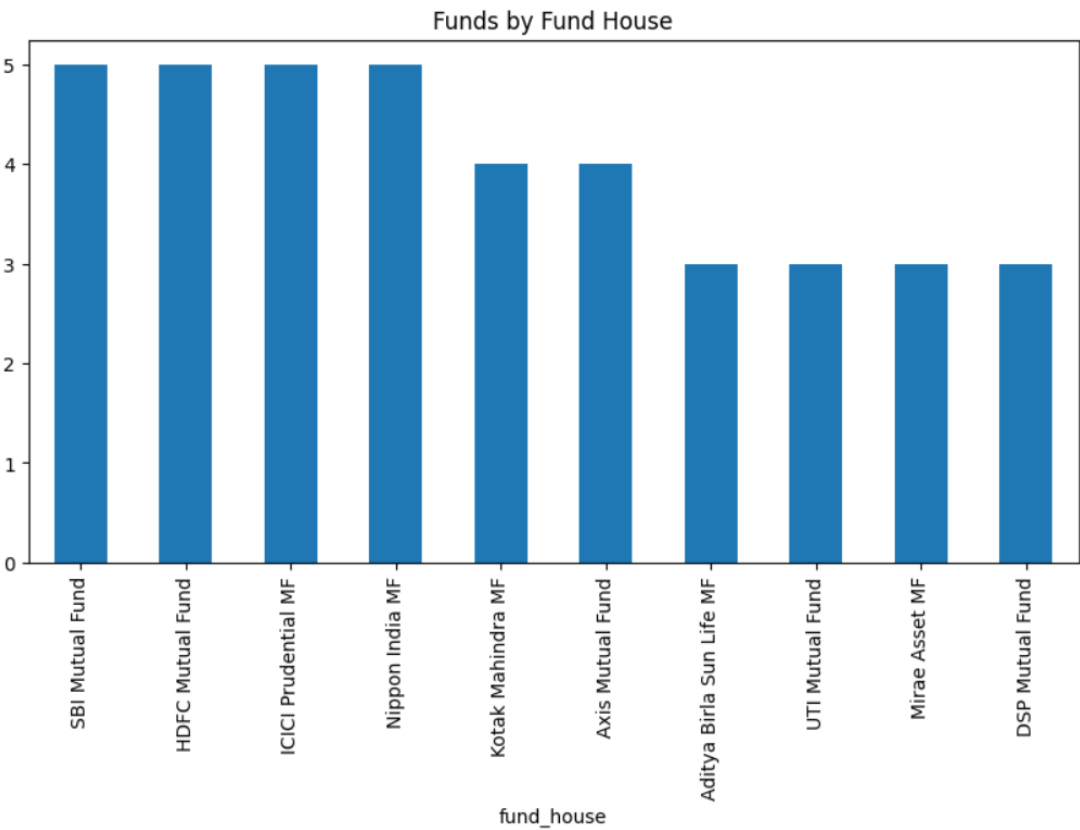
- Faster query execution.
- Improved reporting performance.
- Simplified analytical calculations.

- Better scalability.

EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis

EDA was performed to understand trends and relationships within the data.

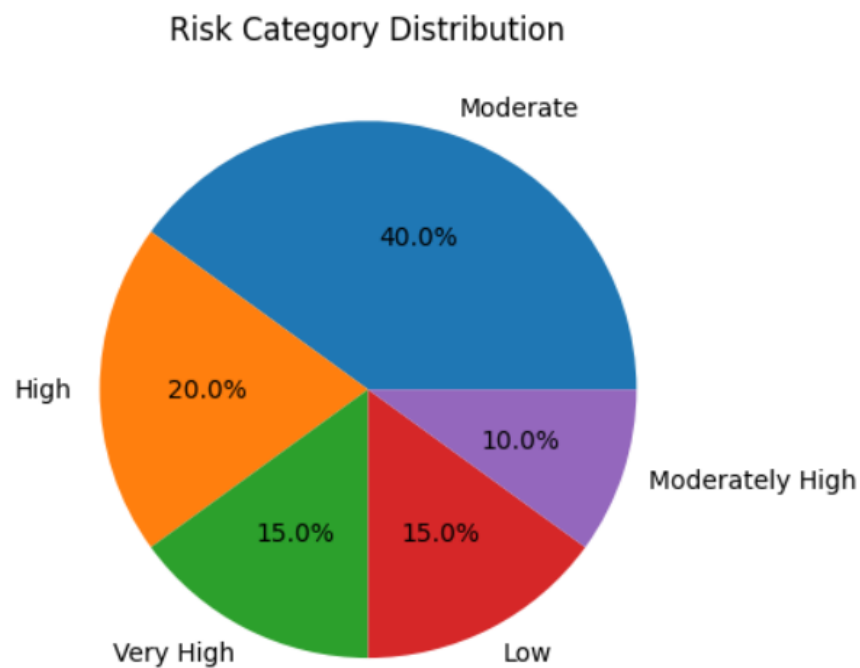


Fund House Distribution

Analysis showed concentration among major AMCs including SBI, HDFC, ICICI Prudential, Nippon India, and Kotak.

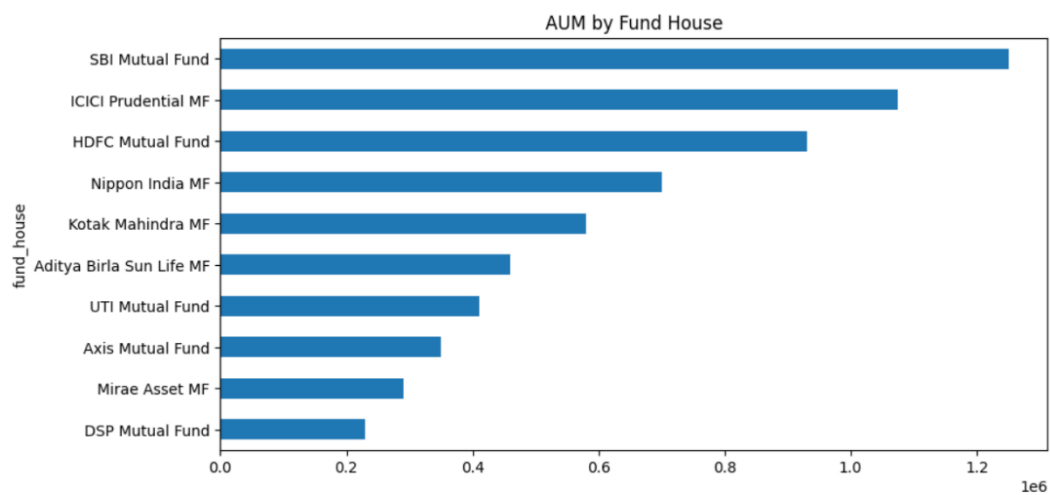
Risk Category Distribution

Most schemes belonged to Moderate to High risk categories.



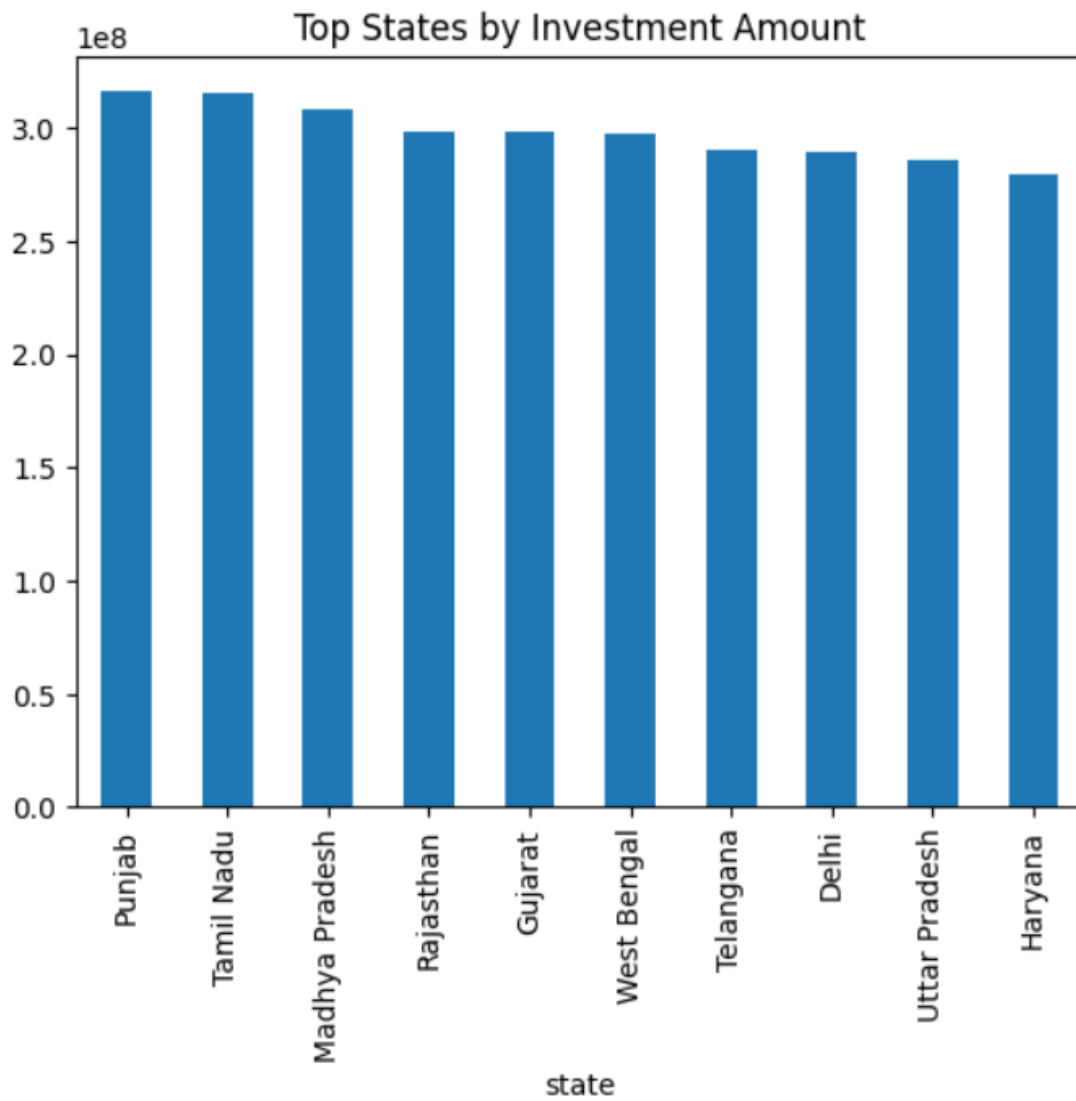
Category Analysis

Large Cap, Flexi Cap, and Small Cap categories dominated the dataset.



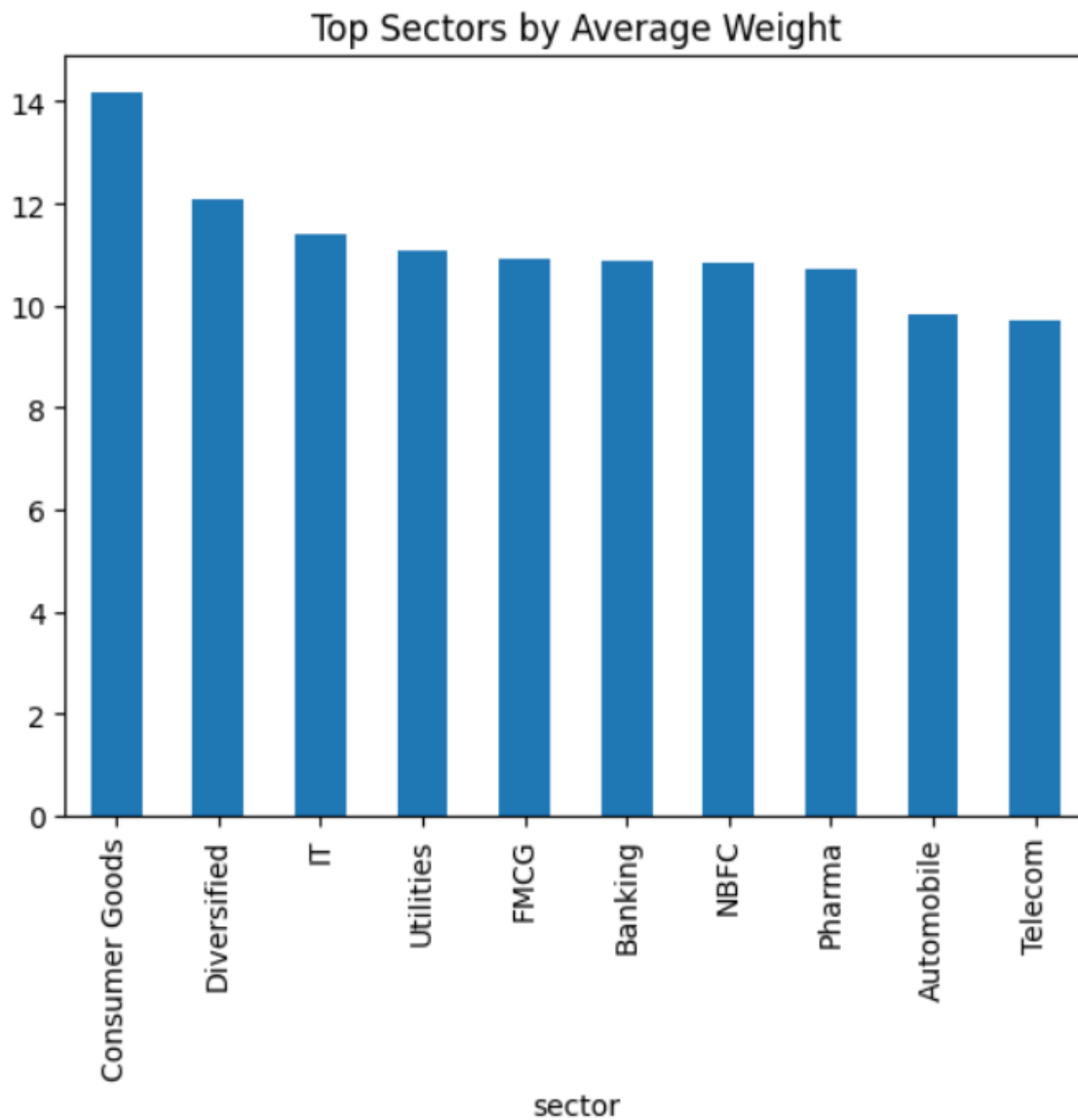
Investor Analysis

Investment activity was concentrated among selected states and metropolitan regions.



Portfolio Analysis

Financial Services and Technology sectors represented significant portfolio allocations.



PERFORMANCE ANALYTICS

Performance Metrics

Several quantitative performance measures were calculated.

CAGR

Measures annualized investment growth.

	amfi_code	CAGR
0	100016	0.026352
1	100025	0.044551
2	100033	0.300997
3	101206	0.235205
4	101207	0.079331

Sharpe Ratio

Measures excess return per unit of total risk.

Sortino Ratio

Measures excess return relative to downside risk.

	amfi_code	Sharpe	Sortino
34	148567	1.448291	2.385644
30	120843	1.306744	2.364320
36	148569	1.234930	2.146914
19	119551	1.208267	2.140267
25	120505	1.180101	2.029353
38	149323	1.132122	1.875101
2	100033	1.093699	1.829134
9	118632	1.081659	1.850133
3	101206	1.027213	1.799563
24	120504	1.026524	1.805294

Alpha

Measures risk-adjusted excess return over benchmark.

Beta

Measures market sensitivity.

	amfi_code	Alpha	Beta
0	100016	0.035273	0.000023
1	100025	0.043864	-0.000056
2	100033	0.268942	0.000175
3	101206	0.212161	0.000137
4	101207	0.104643	0.000128

Maximum Drawdown

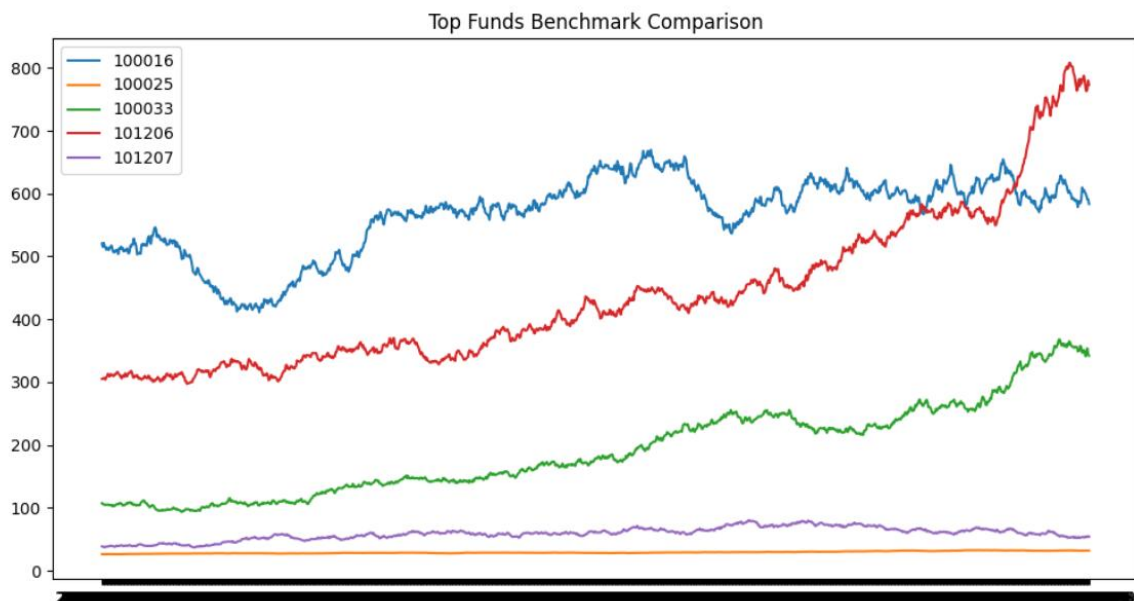
Measures worst portfolio decline from peak value.

	amfi_code	max_drawdown
0	100016	-0.247344
1	100025	-0.043083
2	100033	-0.162172
3	101206	-0.112916
4	101207	-0.354469

Results

Small Cap and Mid Cap funds generated the highest long-term returns but exhibited higher volatility.

	scheme_name	score
0	SBI Small Cap Fund - Regular Plan - Growth	82.7500
1	ICICI Pru Midcap Fund - Regular - Growth	81.6250
2	DSP Small Cap Fund - Regular - Growth	77.3125
3	HDFC Mid-Cap Opportunities Fund - Regular - Gr...	74.8750
4	Kotak Flexicap Fund - Regular - Growth	74.6250
5	Axis Midcap Fund - Regular - Growth	69.6250
6	Mirae Asset Large Cap Fund - Regular - Growth	68.6250
7	DSP Midcap Fund - Regular - Growth	68.3750
8	ICICI Pru Bluechip Fund - Direct - Growth	62.8750
9	Mirae Asset Tax Saver Fund - Regular - Growth	62.0625



ADVANCED ANALYTICS

Value at Risk (VaR)

Historical VaR at 95% confidence level was computed for all schemes.

VaR estimates the maximum expected loss during adverse market conditions.

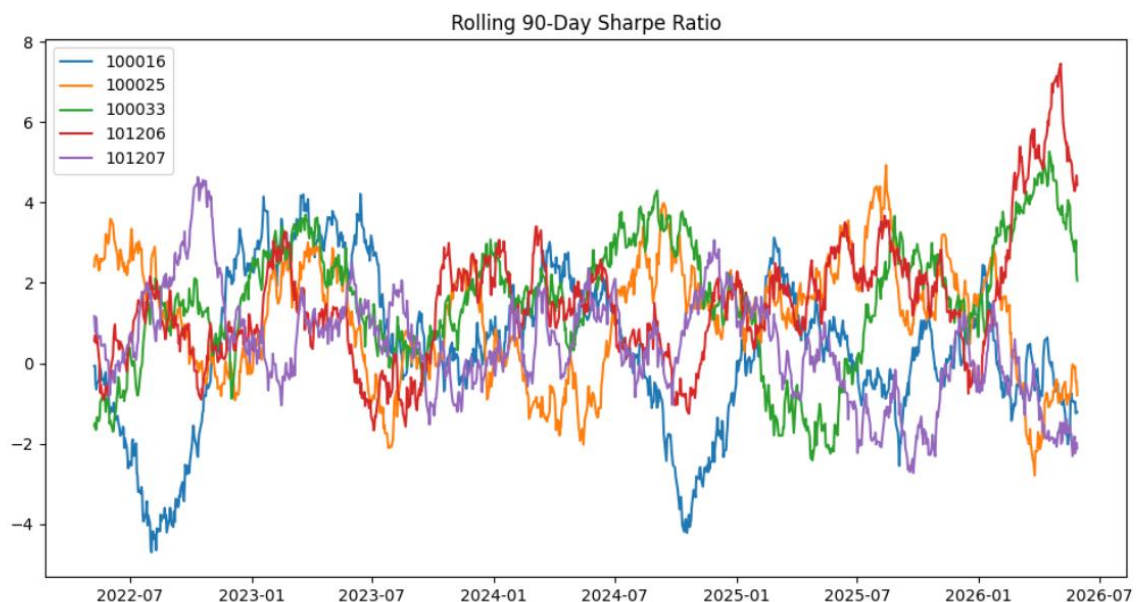
	amfi_code	VaR_95	CVaR_95
0	100016	-0.014364	-0.018060
1	100025	-0.003793	-0.004994
2	100033	-0.019034	-0.023456
3	101206	-0.013282	-0.017439
4	101207	-0.026021	-0.032459

Conditional Value at Risk (CVaR)

CVaR estimates average losses beyond the VaR threshold.

Rolling Sharpe Ratio

A 90-day rolling Sharpe Ratio was calculated to evaluate changing risk-adjusted performance.



Sector Concentration Analysis

Herfindahl-Hirschman Index (HHI) was used to measure portfolio concentration.

	amfi_code	HHI
0	119551	0.118716
1	119552	0.108011
2	119598	0.107349
3	119599	0.174751
4	100016	0.139534

Fund Recommendation System

A recommendation engine was built using:

- Risk Grade
- Sharpe Ratio
- Historical Performance

INVESTOR ANALYTICS

Cohort Analysis

Investors were grouped based on their first investment year.

Metrics analyzed:

	cohort	avg_amount	total_invested
0	2024	107422.541832	3491125187
1	2025	109158.577061	30455243

- Average SIP Amount
- Total Invested Amount
- Preferred Mutual Funds

SIP Continuity Analysis

Investors with six or more SIP transactions were analyzed.

Metrics:

- Average Gap Between SIPs
- Continuity Rate
- At-Risk Investors

	investor_id	avg_gap	status
0	INV000004	85.400000	At-Risk
1	INV000008	70.400000	At-Risk
2	INV000010	64.800000	At-Risk
3	INV000011	40.166667	At-Risk
4	INV000012	57.000000	At-Risk

Transaction Analysis

Transactions were segmented by:

- State
- Age Group
- City Tier
- Transaction Type

Key finding: SIP remains the dominant investment mode among retail investors.

POWER BI DASHBOARD

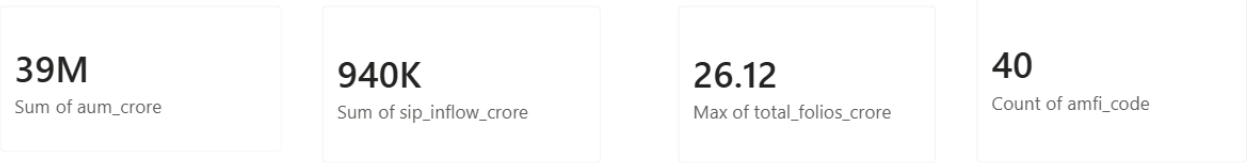
Dashboard Overview

A four-page interactive Power BI dashboard was developed.

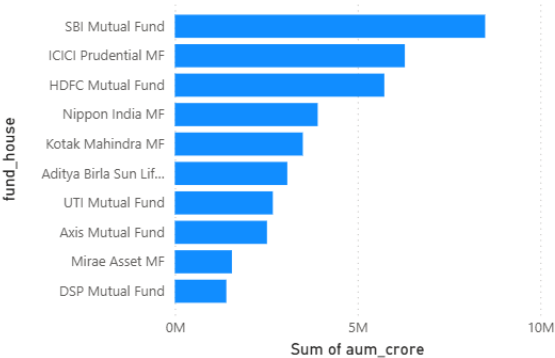
Page 1 – Industry Overview

- Total AUM
- SIP Inflows
- Folio Count
- Industry Trends

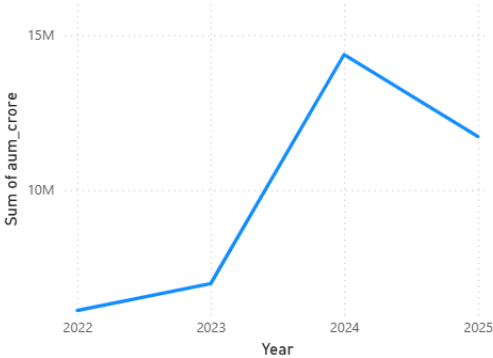
Bluestock Mutual Fund Analytics Dashboard
Indian Mutual Fund Industry Overview (2022–2025)



Sum of aum_crore by fund_house



Sum of aum_crore by Year

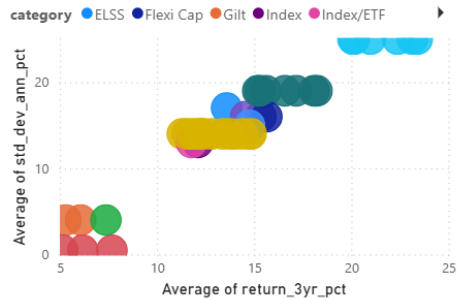


Page 2 – Fund Performance

- Risk vs Return Analysis
- NAV Trends
- Fund Scorecard

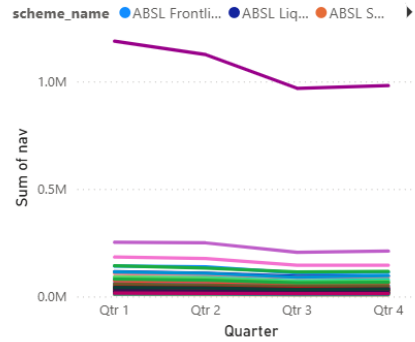
Fund Performance & Risk Analytics

Average of return_3yr_pct, Average of std_dev_ann_pct and Average of aum_crore by scheme_name and category



scheme_name	fund_house
ABSL Frontline Equity Fund - Regular - Growth	Aditya Birla
ABSL Liquid Fund - Regular - Growth	Aditya Birla
ABSL Small Cap Fund - Regular - Growth	Aditya Birla
Axis Bluechip Fund - Direct - Growth	Axis Mutual Fund
Axis Bluechip Fund - Regular - Growth	Axis Mutual Fund
Axis Midcap Fund - Regular - Growth	Axis Mutual Fund
Axis Small Cap Fund - Regular - Growth	Axis Mutual Fund
DSP Midcap Fund - Regular - Growth	DSP Mutual Fund
DSP Small Cap Fund - Regular - Growth	DSP Mutual Fund
DSP Top 100 Equity Fund - Regular - Growth	DSP Mutual Fund
Total	

Fund NAV Trend



fund_house

- ☐ Aditya Birla Sun Life MF
- ☐ Axis Mutual Fund
- ☐ DSP Mutual Fund
- ☐ HDFC Mutual Fund
- ☐ ICICI Prudential MF
- ☐ Kotak Mahindra MF
- ☐ Mirae Asset MF
- ☐ ...

Risk vs Return Analysis

category

☐ Debt

☐ Equity

plan

☐ Direct

☐ Regular

23.39

Max of return_3yr_pct

7.68

Max of sharpe_ratio

1.98

Max of alpha

0.55

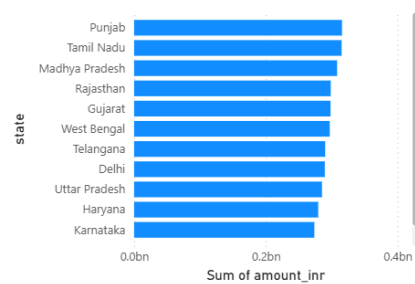
Min of expense_ratio_pct

Page 3 – Investor Analytics

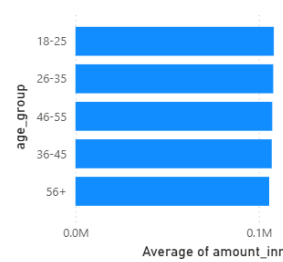
- State Analysis
- Transaction Analysis
- Age Group Analysis

Investor Behavior & Transaction Analytics

Transaction Amount by State



Average Investment by Age Group



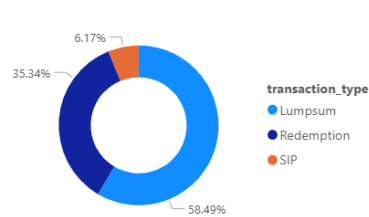
state

- ☐ Delhi
- ☐ Gujarat
- ☐ Haryana
- ☐ Karnataka

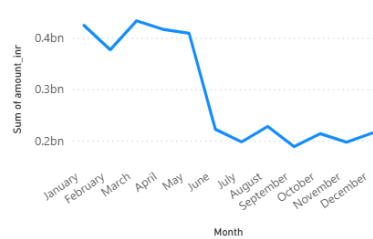
age_group

- ☐ 18-25
- ☐ 26-35
- ☐ 36-45
- ☐ 46-55

Transaction Type Distribution



Monthly Transaction Volume



city_tier

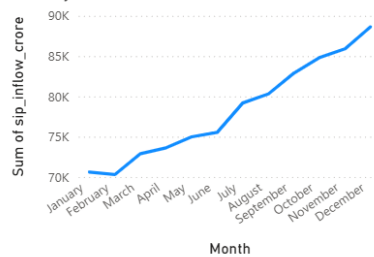
- ☐ B30
- ☐ T30

Page 4 – SIP & Market Trends

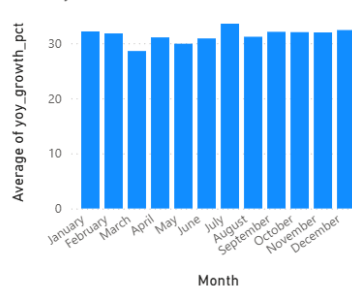
- SIP Growth
- Category Inflows
- Market Trends

SIP Growth & Market Trend Analysis

Monthly SIP Inflow



Monthly SIP YoY Growth (%)



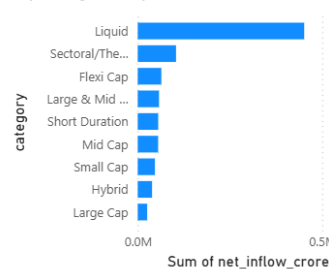
Category-wise Net Inflow Analysis

category	01 April 2024	01 May 2024
ELSS	466	553
Flexi Cap	4947	5529
Gilt	784	836
Hybrid	2955	3487
Large & Mid Cap	4214	4368
Large Cap	2413	2076
Liquid	37537	41872
Mid Cap	3897	5300
Sectoral/Thematic	8052	8354
Short Duration	4400	4833
Small Cap	3533	4092
Total	74526	82661

month

01-04-2024 01-03-2025

Top Categories by Net Inflow



KEY INSIGHTS

Major Findings

1. Mutual fund industry AUM has shown strong growth between 2022 and 2025.
2. SIP inflows continue to increase steadily, indicating rising retail participation.
3. Small Cap funds delivered the highest returns but also carried higher risk.
4. Risk-adjusted metrics identified several funds consistently outperforming benchmarks.
5. Financial Services and Technology sectors dominate portfolio allocations.
6. Investor activity is concentrated in major states and metropolitan areas.
7. Equity-oriented funds attracted the highest net inflows.
8. Portfolio concentration varies significantly across schemes.
9. Rolling Sharpe Ratios revealed changing risk-adjusted performance over time.
10. The recommendation system successfully identifies suitable funds based on investor risk preferences.

CONCLUSION & FUTURE WORK

Conclusion

The Mutual Fund Analytics Platform successfully integrates data engineering, financial analytics, and business intelligence into a single solution.

The project demonstrates how modern analytics techniques can support investment decision making through automated data processing, risk assessment, and interactive visualization.

The developed platform enables investors, analysts, and financial institutions to gain deeper insights into mutual fund performance and investor behavior.

Future Enhancements

1. Real-time NAV streaming.
2. Streamlit web application.
3. Monte Carlo portfolio simulation.
4. Markowitz portfolio optimization.
5. Automated email reporting.
6. Machine learning-based return forecasting.
7. Cloud deployment and API integration.

References

- AMFI India
- MFAPI
- NSE India
- RBI Publications
- Mutual Fund Industry Reports

Thank You